

RTL Design Sherpa

RAPIDS Beats DMA Engine Phase 1 Characterization Report

July 4, 2026

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1 RAPIDS Beats DMA Engine — Phase 1 Characterization Report

Component: RAPIDS beats DMA (`rapids_beats_top`) **Platform:** Digilent Nexys A7-100T (Xilinx Artix-7 100T, 100 MHz, **4-channel build**, -1 speed grade) **Status:** Phase 1 complete — the engine was split into two wholly-separate data paths, characterized on real silicon, timing-closed at 100 MHz, and both paths CRC-validated against a deterministic golden model across a full config sweep.

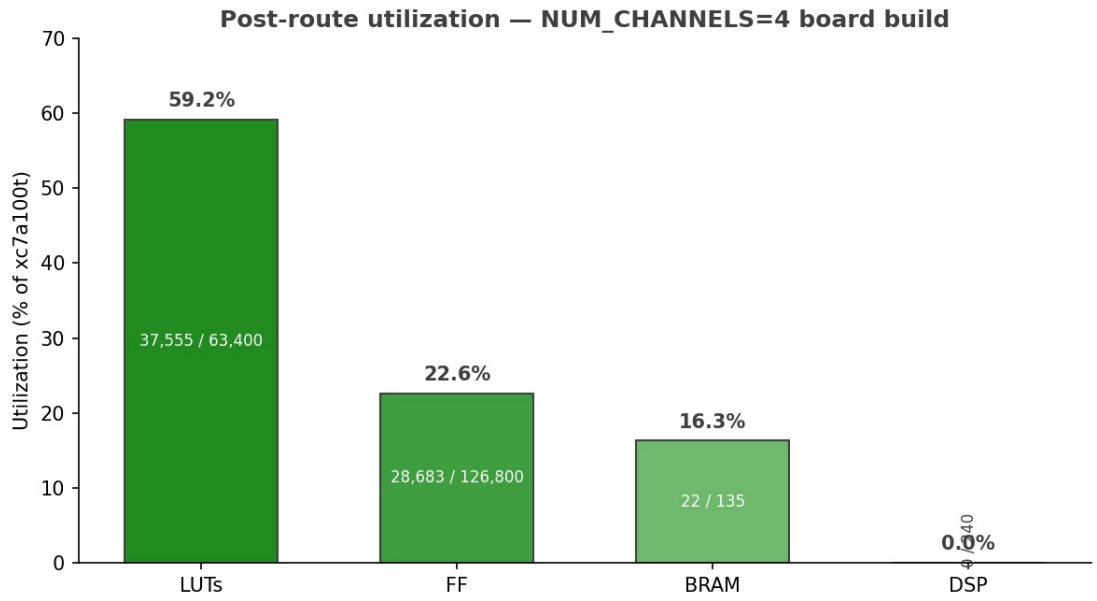
Headline: the RAPIDS beats DMA was refactored into two wholly-separate engines — a write-only sink and a read-only source — driven by a directional scheduler. It was characterized on a Nexys A7-100T, closed timing at 100 MHz with +**0.007 ns of setup slack and zero failing endpoints**, and both data paths were CRC-validated on silicon across a **48/48-PASS** sweep of channels, beat counts, backpressure, and seeds.

1.1 0. Resource footprint

Before any dynamic behavior, the static cost. The numbers below are post-route on `xc7a100tcsg324-1`, **NUM_CHANNELS** = **4**, `SRAM_DEPTH = 256`, `DESC_RAM_ENTRIES = 256`, all monitors enabled, harness included.

Resource	Used	Available	Utilization
Slice LUTs	37,555	63,400	**59.2 %**
Slice Registers (FF)	28,683	126,800	**22.6 %**
Block RAM tiles	22	135	**16.3 %**
DSP	0	240	**0 %**

1.1.1 Figure 0.1: Post-route utilization — 4-channel board build



post-route utilization

LUTs are the tight resource at 59.2 %; the two SRAMs plus the two descriptor RAMs plus the harness pattern-memory account for the BRAM. There is no DSP in the design — the beats engine is pure control and datapath movement, no arithmetic reduction.

1.1.2 0.1 The BRAM board-fit trim

The harness ships with generous simulation defaults — `SRAM_DEPTH = 4096` and `DESC_RAM_ENTRIES = 2048` — chosen so the cocotb testbench never has to think about buffer

pressure. On the 100T those defaults are not synthesizable as-is: the SRAMs and descriptor RAMs inflate to **148 BRAM tiles**, which overflows the 135-tile device. For the board build the depths were trimmed to 256 each, which drops the footprint to **22 tiles (16.3 %)** with comfortable headroom. This is a build-time parameter only; the RTL is unchanged, and the sweep in Section 9 confirms the trimmed depths cover every characterized transfer size.

1.2 1. What we set out to do

RAPIDS is a descriptor-driven accelerator DMA. The beats variant moves fixed “beat” quanta between an AXIS network interface and system memory. Before declaring the split-engine refactor ready, we wanted to **measure** what it actually does on real silicon — not just pass simulation. Specifically:

1. **Prove the split is clean.** The engine was refactored from a single bidirectional block into two wholly-separate engines — a write-only sink and a read-only source. Does the split hold on silicon, with no hidden coupling between the two paths?
2. **Close timing at 100 MHz on the -1 part.** The shared monitor-bus core carries a config-CSR path that had been the chronic critical path on this family. Can we close it without pipelining the datapath?
3. **Validate both data paths end-to-end.** Sink (network to memory) and source (memory to network) each need independent, deterministic correctness checking that survives multi-channel interleave.
4. **Establish a reproducible on-silicon regression.** A smoke test and a full config sweep, driven over UART, with machine-readable results — the same flow shape STREAM already uses.

The headline answer: the split is clean, timing closes with `+0.007 ns` of slack, and both paths CRC-match a golden model across all 48 swept configurations. The rest of this document is the work behind those results.

1.3 2. Test platform

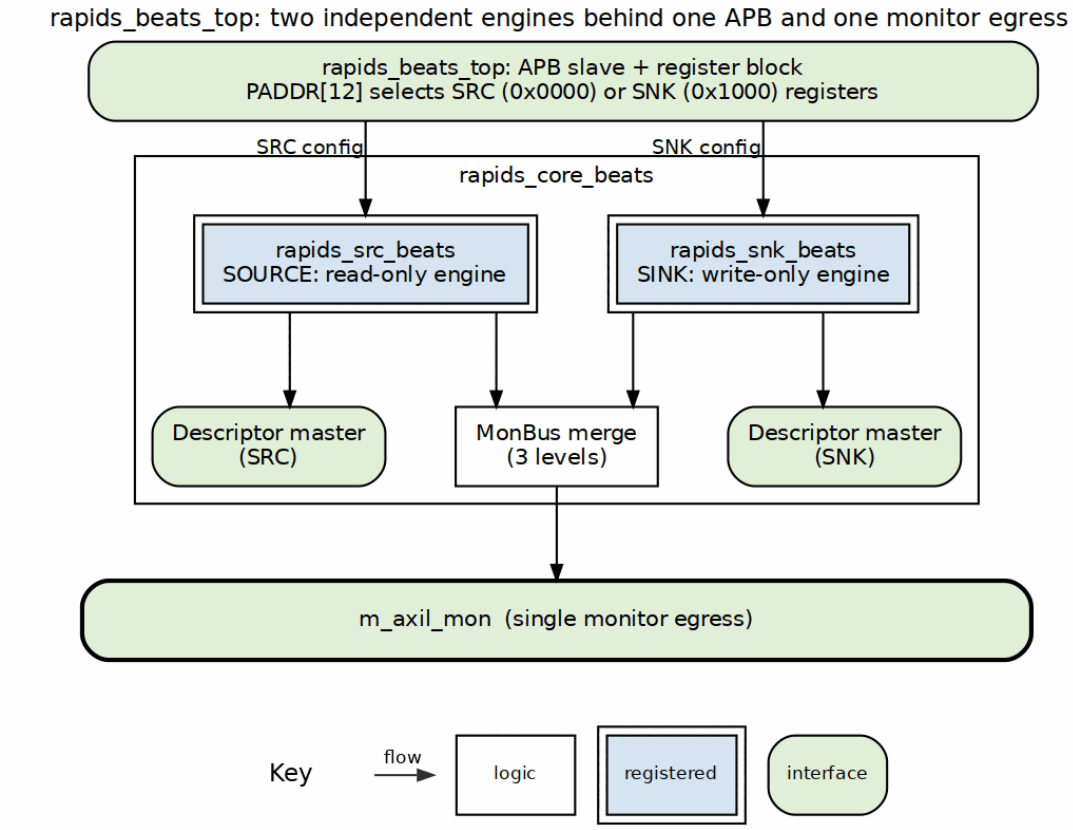
FPGA	Xilinx Artix-7 100T (`xc7a100tcsg324-1`, -1 speed grade)
Clock	`aclk` = 100 MHz (10 ns period)
Build	`rapids_char_top` wrapping `rapids_beats_top`, NUM_CHANNELS = 4
Buffer depths (board)	`SRAM_DEPTH = 256`, `DESC_RAM_ENTRIES = 256`
Network interface	AXIS4 (`s_axis` sink in, `m_axis` source out), `tid` = channel
Register map	one APB slave, SRC @ `0x0000`, SNK @ `0x1000`
Monitor egress	single `m_axil_mon` after a 3-level monbus merge
Post-route timing	setup WNS +**0.007 ns**, hold WHS +0.011 ns, 0 failing endpoints
Bitstream resources	37.6k LUTs (59.2 %), 28.7k FFs (22.6 %), 22 BRAM (16.3 %), 0 DSP
Host link	UART, board ID `"RAP1"` (`0x52415031`)

The DUT is `rapids_beats_top`. Everything around it is characterization harness — not part of the IP, but instrumented to make the engine measurable and self-checking on real silicon.

1.4 3. Architecture — two wholly-separate engines

The defining change in this generation is that the beats DMA is no longer one bidirectional block. It is two engines that share nothing but a register decode and a monitor egress.

1.4.1 Figure 3.1: Split architecture



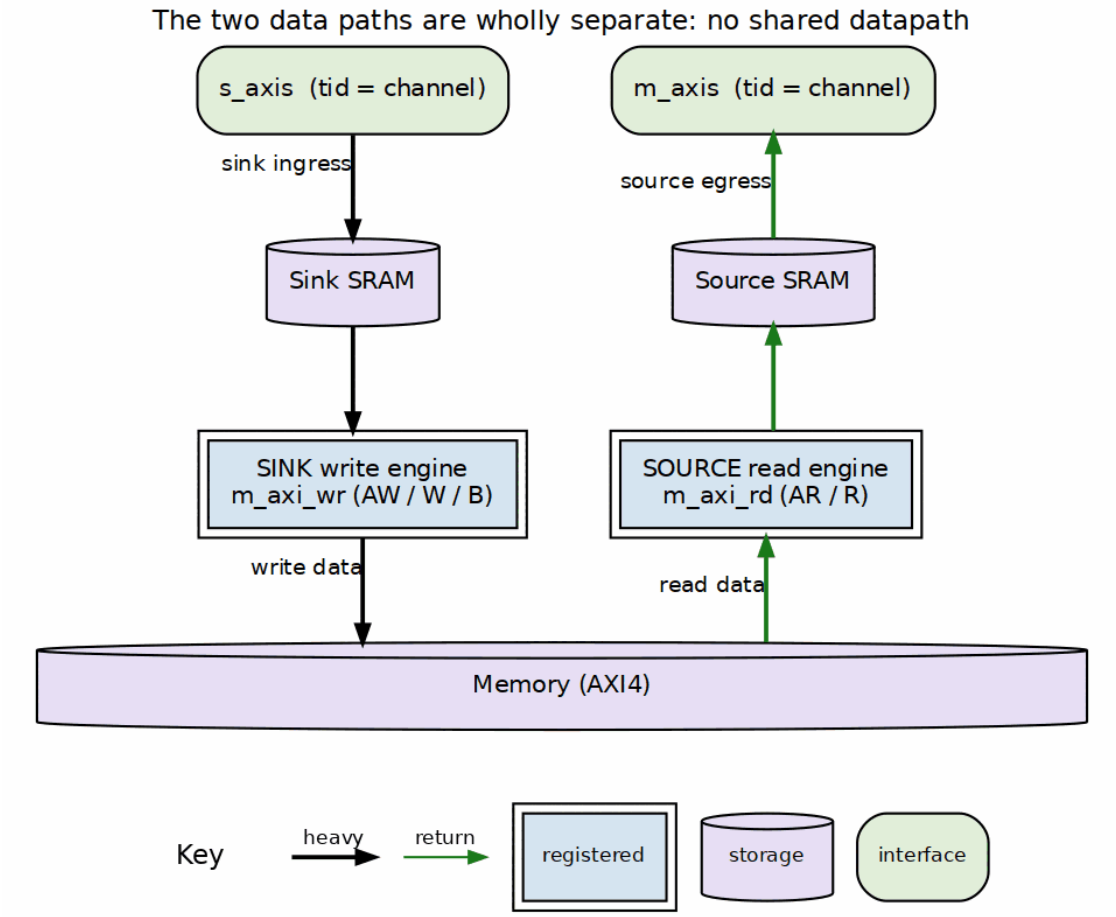
split architecture

`rapids_beats_top` instantiates `rapids_core_beats`, which contains exactly two data engines:

- **`rapids_snk_beats`** — write-only. It receives from the network and writes to memory. It never issues an AXI read.
- **`rapids_src_beats`** — read-only. It reads from memory and sends to the network. It never issues an AXI write.

The two are driven by a **directional scheduler** parameterized by `EN_READ` / `EN_WRITE`, so each engine is compiled with only the half of the scheduler it needs. A single APB decode fans configuration to both halves — **SRC at `0x0000`, SNK at `0x1000`** — and each engine drives its own descriptor master. Monitoring from both halves is merged down to a single `m_axil_mon` egress.

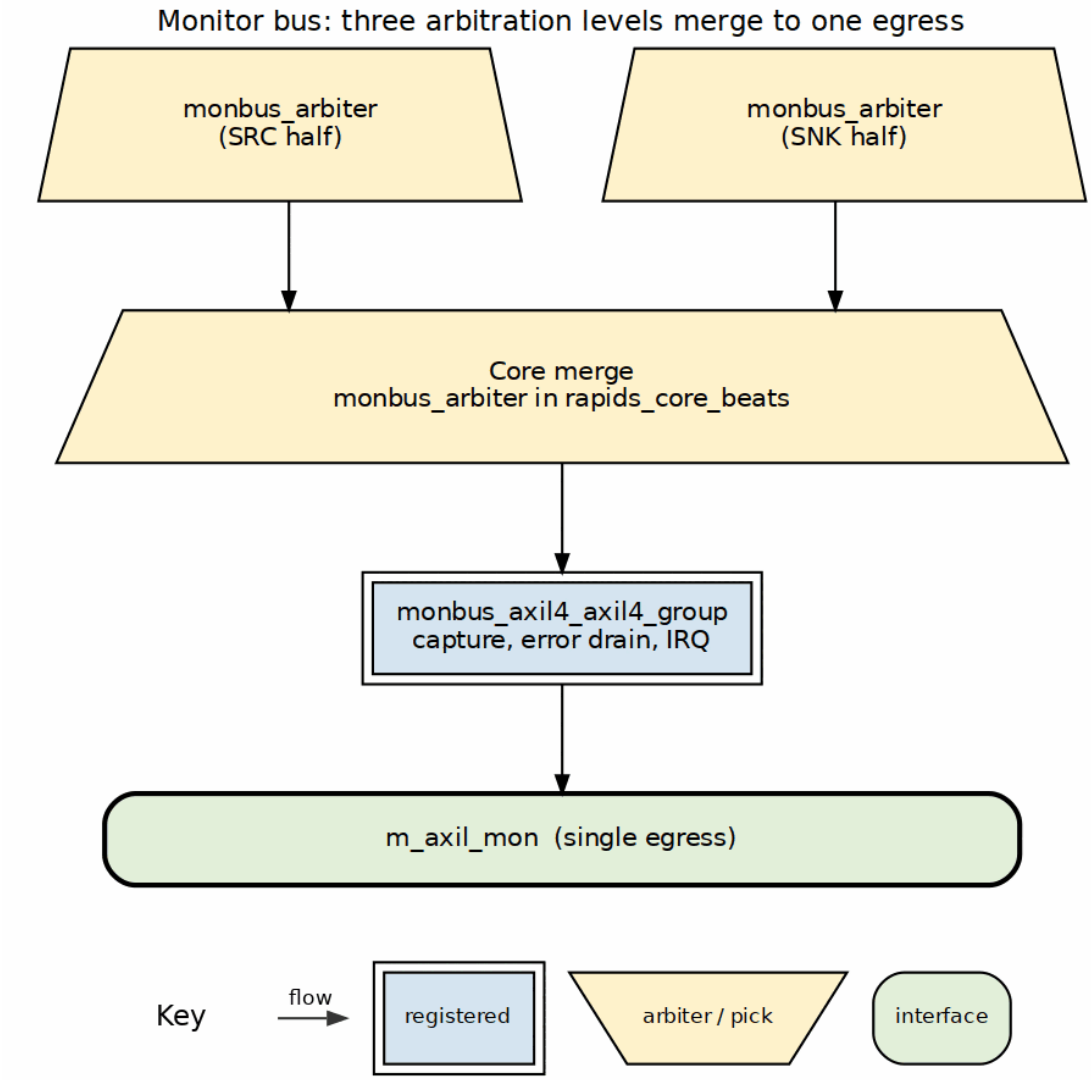
1.4.2 Figure 3.2: The two data paths are wholly separate



data paths

The sink path is `s_axis` (with `tid` = channel) into the sink SRAM, then out `m_axi_wr` (AW/W/B) to memory. The source path is memory in via `m_axi_rd` (AR/R), into the source SRAM, then out `m_axis` (with `tid` = channel). There is no shared datapath, no shared buffer, and no shared AXI master between the two directions. This is what “wholly separate” means in practice: a change to one engine cannot introduce a datapath hazard in the other.

1.4.3 Figure 3.3: Monitor-bus hierarchy — 3-level merge to one egress



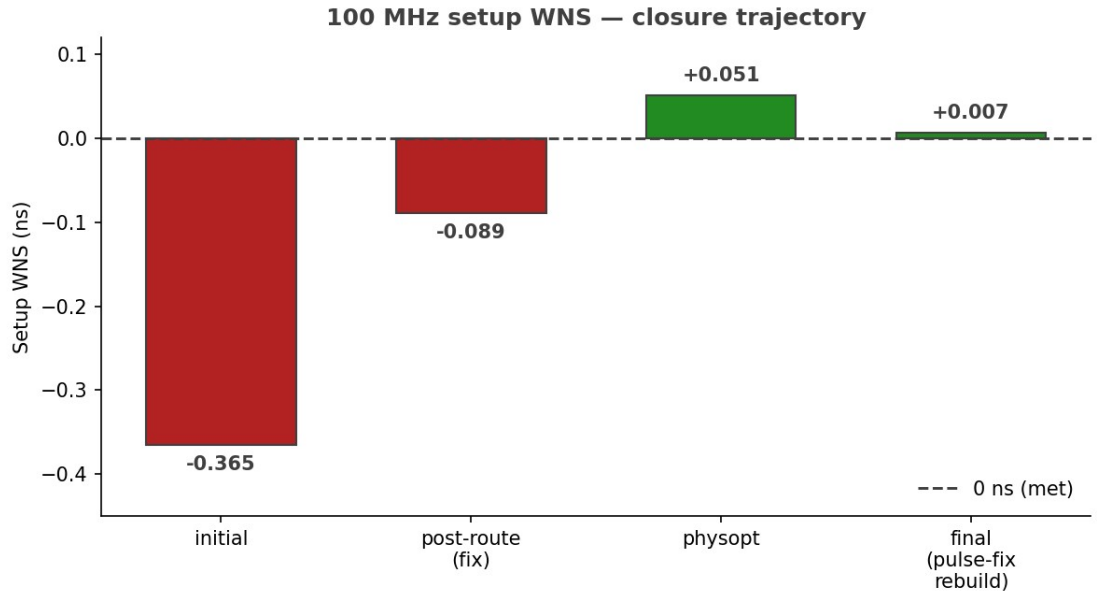
monbus hierarchy

Observability is built in and merges in three levels: a per-half `monbus_arbiter` inside each of the source and sink engines, a core-level `monbus_arbiter` that merges the two halves, and finally `monbus_axil4_axil4_group`, which drains the merged stream to the single `m_axil_mon` egress. One wire off-chip carries every monitor packet from both directions.

1.5 4. Timing closure at 100 MHz

Post-route, the build closes with **setup WNS** = +**0.007 ns, 0 failing endpoints**, and hold met (WHS = +0.011 ns). Getting there was not free.

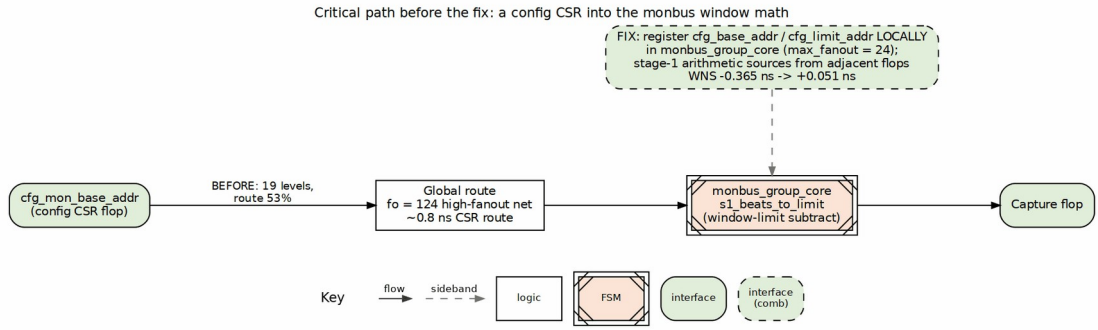
1.5.1 Figure 4.1: 100 MHz setup WNS — closure trajectory



WNS closure trajectory

The closure trajectory ran through four points: an initial post-route WNS of **-0.365 ns** (failing), **-0.089 ns** after the fix landed at post-route, +**0.051 ns** after phys-opt, and a final +**0.007 ns** on the rebuild that also folded in the harness pulse-fix of Section 8.

1.5.2 Figure 4.2: The critical path — a config CSR into the monbus window math



critical path

The critical path was **not** in the datapath. It ran from `cfg_mon_base_addr` — a configuration CSR that essentially never changes at runtime — into `monbus_group_core`'s `s1_beats_to_limit`, the stage-1 window-limit subtract. At 19 logic levels it looked like a logic-depth problem, but it was **route-dominated**: route delay was **53 %** of the path, including a high-fanout net (`fo = 124`) and roughly **0.8 ns** of pure CSR route just to reach the arithmetic.

The fix was to **register `cfg_base_addr` / `cfg_limit_addr` locally inside `monbus_group_core`** with `max_fanout = 24`, so the stage-1 subtract sources its operands from flops physically adjacent to the arithmetic rather than from a CSR on the far side of the die. That change alone moved the endpoint from **-0.365 ns to +0.051 ns** — no datapath pipelining required. Because `monbus_group_core` is shared collateral, **STREAM benefits from the same fix**.

1.6 5. Vivado bring-up fixes (found vs. Verilator)

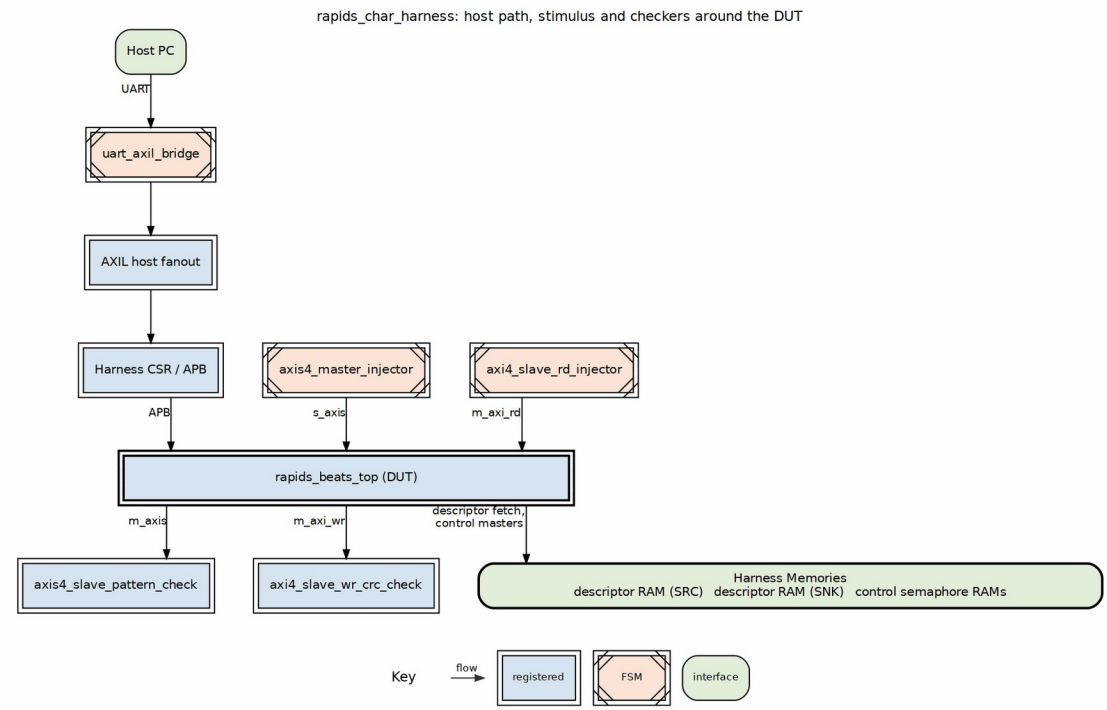
The design simulated cleanly under Verilator but Vivado elaboration surfaced a class of issue Verilator does not: package-symbol collisions under `$unit` wildcard imports. Vivado merges all `import pkg::*;` into one `$unit` scope, and guarded imports only apply to the first file that imports them — so a symbol defined in two packages becomes ambiguous the moment a second file pulls both in. The fixes, all in the RTL's package qualification (no logic change):

- **`rapids_pkg` vs. `stream_pkg` enum-label collisions.** Both packages define `RD_*` and `CH_*` enum labels. Ambiguous references were qualified with `rapids_pkg::`.
- **Type collisions.** `channel_state_t`, `descriptor_t`, and `read_engine_state_t` exist in both packages; each use site was qualified with `rapids_pkg::`.
- **Monitor response codes.** `ocelus_amba4_pkg::AXI_ERR_RESP_*` was qualified in the control engines, where it otherwise clashed under the merged `$unit` scope.

None of these are visible under Verilator's per-file import handling; they only appear once Vivado flattens the import scope. They are documented here so the next engine that reuses these packages qualifies from the start.

1.7 6. The characterization harness

1.7.1 Figure 6.1: `rapids_char_harness` block diagram



harness block diagram

The harness wraps `rapids_beats_top` with everything needed to drive it from a host PC and check both directions on-chip, while leaving the DUT untouched:

- **On-chip AXIS pattern generator and checker.** `axis4_master_injector` drives `s_axis` for the sink path; `axis4_slave_pattern_check` consumes `m_axis` on the source path. Both are multi-channel with a **per-channel LFSR seeded `seed ^ ch`** and a **CRC-32** computed over the actual beats, so the data is deterministic and independently reproducible per channel.
- **Memory-side pattern / CRC.** `axi4_slave_rd_injector` backs `m_axi_rd` with the same deterministic LFSR stream (this is the source path's memory), and `axi4_slave_wr_crc_check` CRCs everything written on `m_axi_wr` (the sink path's memory).
- **Dual descriptor RAMs.** One descriptor RAM per engine, feeding the two independent descriptor masters.
- **Semaphore RAMs** on the control masters, backing the engines' control-plane accesses.

- **Host link.** A `uart_axil_bridge` fans the single UART into the AXIL host bus, reaching the harness CSRs and the DUT's APB. The board announces itself with ID `"RAP1"` (`0x52415031`) so the host script can confirm it is talking to the right bitstream.

Every block earned its place: the split engines mean the sink and source each need their own traffic source and their own checker, and the per-channel LFSR / CRC keeps correctness meaningful under multi-channel `tid` interleave.

1.8 7. CRC validation on both paths against a golden model

Both directions are checked against a **deterministic software golden model**, `flows-rapids-beats/host/rapids_char_golden.py`, which computes the **reflected CRC-32** over the per-channel LFSR stream. The model is the single source of truth: the same expected value is asserted on the write side (sink), the read side (source), and the on-chip checker.

For an 8-beat transfer the golden model reproduces, per channel:

Channel	Golden CRC-32
ch0	`0x8C023372`
ch1	`0x3FB81189`
ch2	`0xD64B4EEC`
ch3	`0x65F16C17`

These were validated in two independent places:

- **In simulation** — the cocotb harness testbench asserts that the sink write CRC, the source read CRC, and the on-chip checker CRC all equal the golden value (2/2 checks pass).
- **On the host** — `rapids_char_golden.py` reproduces the same per-channel values standalone, so a mismatch on hardware points at the RTL, not at a disagreement between the checker and the model.

Because the golden model is deterministic in `(seed, channel, beats)`, the sweep in Section 9 can assert exact per-channel CRC equality for every one of its 48 configurations rather than just "no error flagged."

1.9 8. A real harness bug, root-caused

The most instructive find of the campaign was **only visible on hardware**, not in simulation.

The AXIS pattern generator's `start` control was implemented as a **held level** — assert the CSR bit, the generator runs. In simulation that is harmless: the testbench edge-arms the generator and resets it between runs, so the level is asserted for exactly one controlled window. Over UART it is not harmless. A single host write holds that level for the **~1–2 ms** it takes the next UART transaction to arrive, and the generator **re-arms and re-runs repeatedly** for the entire duration of the held level. Each spurious re-run pushes more beats into the sink than the checker expects, and the sink **desyncs** — a failure that never reproduced in sim because sim never holds the level across real UART timing.

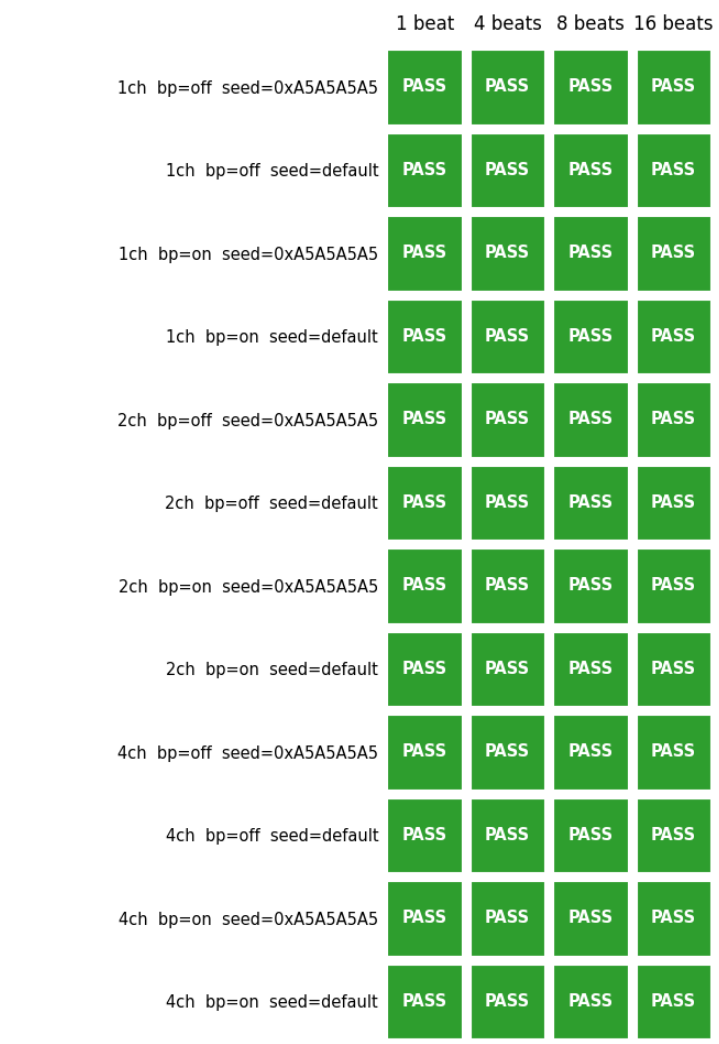
The fix was to make the generator and checker `start` controls **one-cycle pulses** (the same pattern already used by `cam_clear` elsewhere in the harness) rather than held levels. One arm equals exactly one run, independent of how long the host leaves the CSR bit set. This is a textbook example of a bug that a simulation-only flow cannot catch: it lives entirely in the interaction between a held CSR level and real-world host timing.

1.10 9. On-silicon results

The engine ships with two on-silicon regression targets driven over UART, both producing machine-readable JSON results.

- **`make smoke`** — a quick both-paths run, golden-validated. **PASS.**
- **`make suite`** — the full sweep. **48/48 PASS**, both paths CRC-verified against the golden model on every configuration.

1.10.1 Figure 9.1: On-silicon suite — 48/48 configurations PASS



suite result grid

The suite sweeps a clean cross product:

Axis	Values
Active channels	1, 2, 4
Beats per transfer	1, 4, 8, 16
Source backpressure	off, on
Base seed	default (`0xDEADBEEF`), `0xA5A5A5A5`

That is `3 x 4 x 2 x 2 = 48` configurations. Every cell in Figure 9.1 is green: for each configuration, both the sink path (network to memory) and the source path (memory to network) were exercised, and the per-channel write, read, and generator/checker CRCs were compared against the golden model. There were **zero mismatches and zero errors** across the entire grid.

1.11 10. Tooling

The RAPIDS beats flow deliberately mirrors STREAM's characterization flow so the two engines share operator muscle memory and result tooling:

- **`make smoke`** / **`make suite`** drive the board over UART and emit timestamped JSON (`reports/rapids_char_suite_*.json`) with per-configuration, per-channel golden/measured CRCs and pass flags.
- **`flows-rapids-beats/host/rapids_char_golden.py`** is the standalone golden model, usable independently of the board for offline expected-value generation.
- The **DOCX/PDF report pipeline** (this document) uses the same house `md_to_docx.py` recipe and corporate style sheet STREAM uses, so both characterization reports render identically.

1.12 11. What we learned

1. **The split is clean on silicon.** Two wholly-separate engines — write-only sink, read-only source — with no shared datapath, driven by one directional scheduler and merged only at the monitor egress. The 48/48 sweep confirms neither path perturbs the other.

2. **The chronic critical path was route, not logic.** The `cfg_mon_base_addr` into `s1_beats_to_limit` path looked like 19 levels of logic but was 53 % route. Registering the config operands locally in `monbus_group_core` (`max_fanout = 24`) closed it — from `-0.365 ns` to `+0.051 ns` — with no datapath pipelining, and the fix carries over to STREAM.

3. **Timing closes at 100 MHz with margin to spare in the datapath.** Final setup WNS `+0.007 ns`, hold met, zero failing endpoints on the -1 part.

4. **Vivado catches package collisions Verilator hides.** `rapids_pkg` / `stream_pkg` enum-label and type clashes only surface once Vivado flattens `$unit` imports; qualifying at the use site is the durable fix.

5. **Both data paths are golden-validated, not just error-free.** A deterministic reflected-CRC-32 model checks exact per-channel values on the sink write side, the source read side, and the on-chip checker — in sim and on the host.

6. **Some bugs only exist on hardware.** The held-level AXIS start CSR re-ran the generator for the whole UART inter-transaction gap and desynced the sink; converting `start` to a one-cycle pulse fixed it. A simulation-only flow would never have seen it.